

Surface Crack Detection

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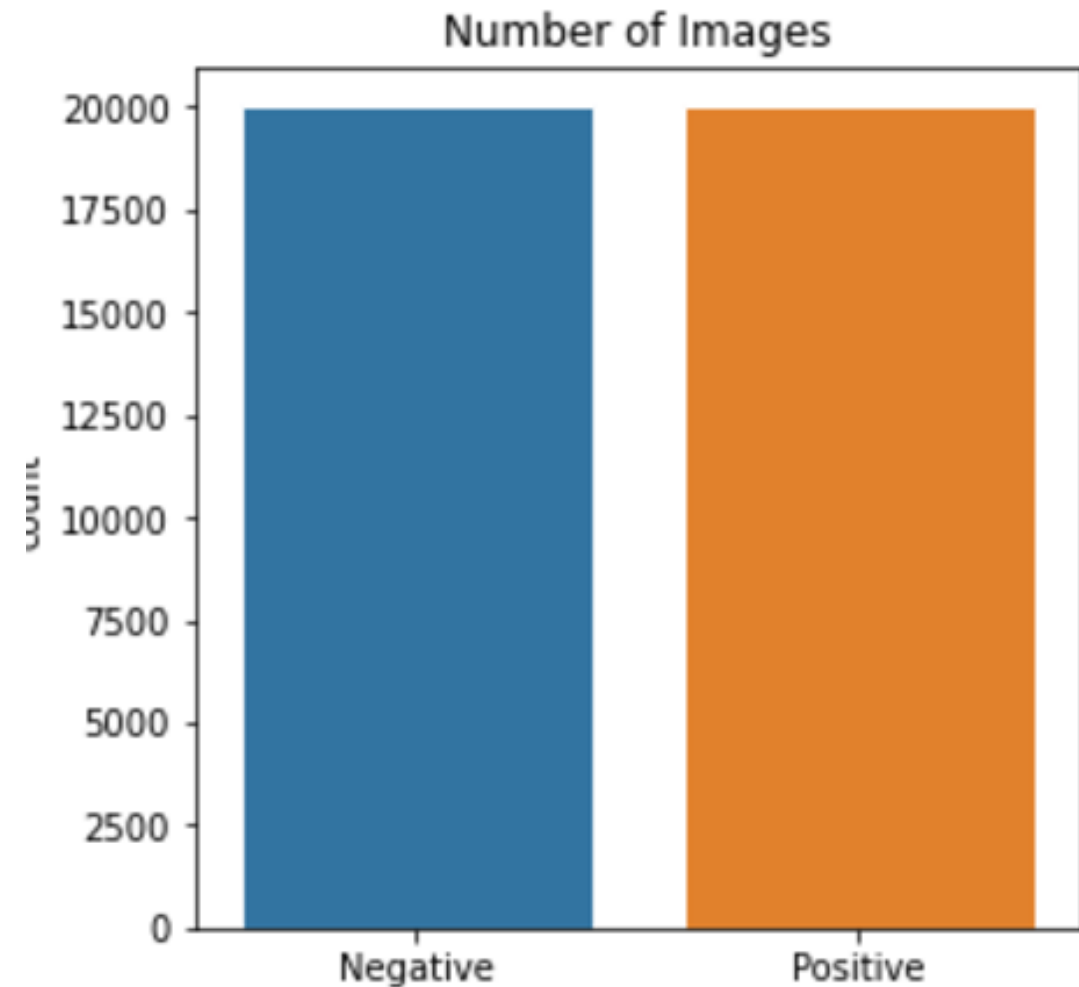
Goal of Study

- The goal of the study is to develop an efficient system for detecting surface cracks in images using convolutional neural networks (CNNs).
- This system aims to address critical issues in industries such as manufacturing and construction by preventing structural failures caused by undetected cracks.
- Additionally, it focuses on leveraging PySpark for large-scale data processing across multiple virtual machines, ensuring scalability and efficiency.



Data Structure

- We collected the data from '**Kaggle**'
- **Total Number of Images:** 40,000 images
- **Classes:** The dataset is divided into **2 classes**:
- **Positive Crack Images:** 20,000 images
- **Negative Crack Images:** 20,000 images



Data Splitting

- **Training Set:** 70% (28,000 images)
- **Validation Set:** 15% (6,000 images)
- **Test Set:** 15% (6,000 images)

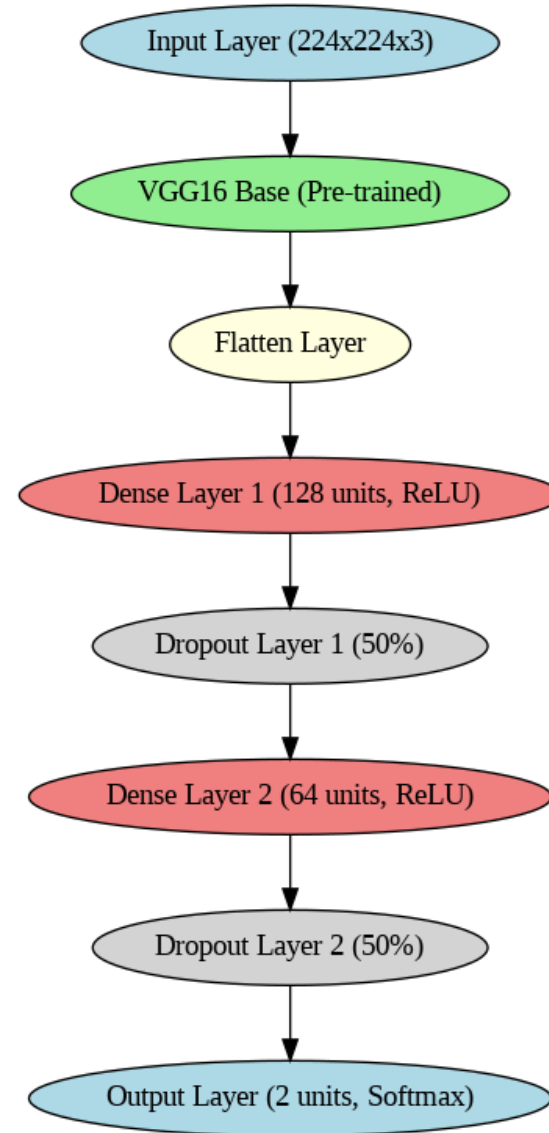




Models

- VGG16
 - CNN
 - ResNet50
 - LeNet
-

VGG 16



Model: "functional_1"

Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 34, 34, 3)	0
block1_conv1 (Conv2D)	(None, 34, 34, 64)	1,792
block1_conv2 (Conv2D)	(None, 34, 34, 64)	36,928
block1_pool (MaxPooling2D)	(None, 17, 17, 64)	0
block2_conv1 (Conv2D)	(None, 17, 17, 128)	73,856
block2_conv2 (Conv2D)	(None, 17, 17, 128)	147,584
block2_pool (MaxPooling2D)	(None, 8, 8, 128)	0
block3_conv1 (Conv2D)	(None, 8, 8, 256)	295,168
block3_conv2 (Conv2D)	(None, 8, 8, 256)	590,080
block3_conv3 (Conv2D)	(None, 8, 8, 256)	590,080
block3_pool (MaxPooling2D)	(None, 4, 4, 256)	0
block4_conv1 (Conv2D)	(None, 4, 4, 512)	1,180,160
block4_conv2 (Conv2D)	(None, 4, 4, 512)	2,359,808
block4_conv3 (Conv2D)	(None, 4, 4, 512)	2,359,808

block3_pool (MaxPooling2D)	(None, 4, 4, 256)	0
block4_conv1 (Conv2D)	(None, 4, 4, 512)	1,180,160
block4_conv2 (Conv2D)	(None, 4, 4, 512)	2,359,808
block4_conv3 (Conv2D)	(None, 4, 4, 512)	2,359,808
block4_pool (MaxPooling2D)	(None, 2, 2, 512)	0
block5_conv1 (Conv2D)	(None, 2, 2, 512)	2,359,808
block5_conv2 (Conv2D)	(None, 2, 2, 512)	2,359,808
block5_conv3 (Conv2D)	(None, 2, 2, 512)	2,359,808
block5_pool (MaxPooling2D)	(None, 1, 1, 512)	0
flatten_1 (Flatten)	(None, 512)	0
dense_3 (Dense)	(None, 128)	65,664
dropout_2 (Dropout)	(None, 128)	0
dense_4 (Dense)	(None, 64)	8,256
dropout_3 (Dropout)	(None, 64)	0
dense_5 (Dense)	(None, 2)	130

Total params: 14,788,738 (56.41 MB)
Trainable params: 74,050 (289.26 KB)
Non-trainable params: 14,714,688 (56.13 MB)



Hyper Parameters

- **Learning Rate:** Start at 0.0001 and use scheduling to decrease it over time for smoother convergence.
- **Grid Search:** Explore different combinations of learning rate, dropout rate, and batch size to find the best configuration.
- **Batch Size:** A batch size of 32 will be used to balance training time and memory usage.
- **Optimizer:** Apply Adam for adaptive learning rates, ensuring faster and more stable convergence.
- **Dropout Rate:** Apply a dropout rate of 0.5 to prevent overfitting during training

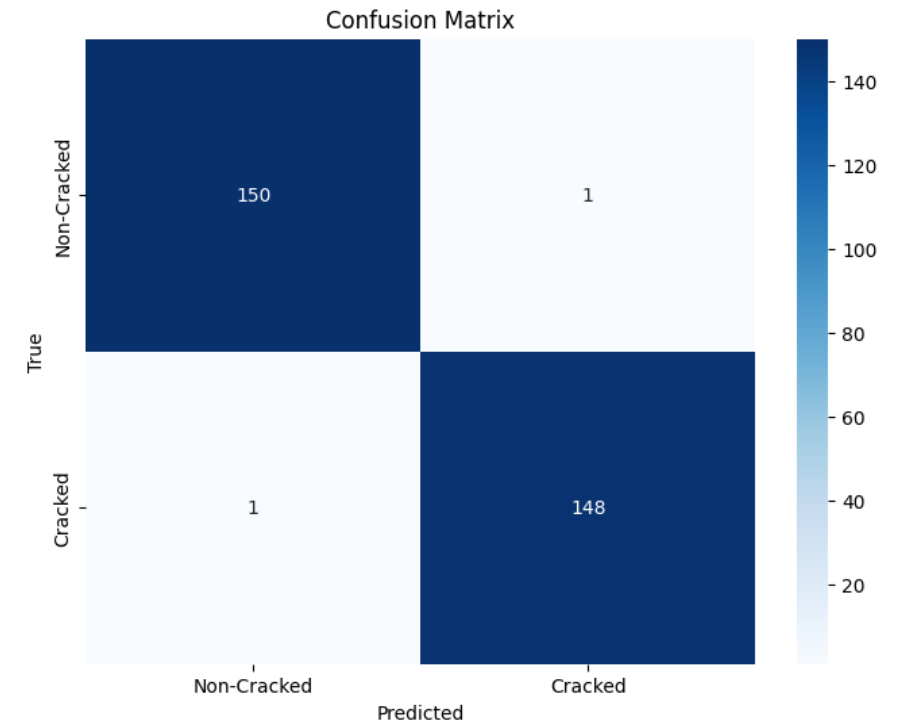


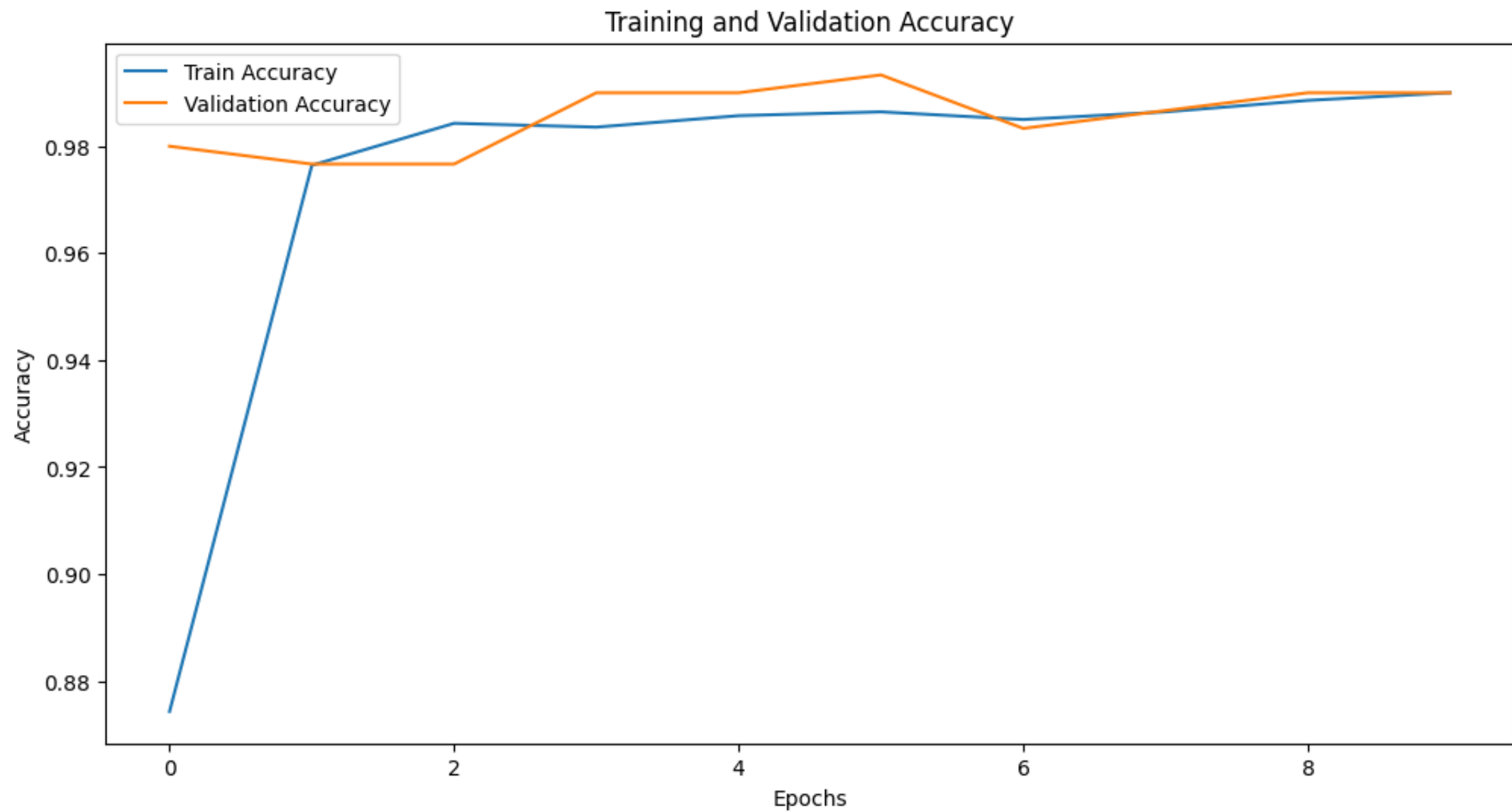
Metrics

```
Classification Report:
              precision    recall  f1-score   support

 Non-Cracked      0.99      0.99      0.99       151
    Cracked      0.99      0.99      0.99       149

 accuracy              0.99              300
 macro avg      0.99      0.99      0.99       300
 weighted avg   0.99      0.99      0.99       300
```





VGG 16

Computational Time

mmagani - hadoop1 - VMware Remote Console

VMRC ▾ ▮ ▮ ▮ ▮

Activities **Firefox** Dec 10 18:58 •

Spark Master at spark://h: × +

← → ↻ 🔒 hadoop1:8080

🌐 http://hadoop1:9870 🌐 http://hadoop1:9864 🌐 spark:4040 🌐 Spark Master 8080

 **Spark Master at spark://hadoop1:7077**

URL: spark://hadoop1:7077
Alive Workers: 2
Cores in use: 16 Total, 0 Used
Memory in use: 13.5 GiB Total, 0.0 B Used
Resources in use:
Applications: 0 [Running](#), 1 [Completed](#)
Drivers: 0 Running, 0 Completed
Status: ALIVE

▾ Workers (2)

Worker Id	Address	State	Cores	Memory	Resources
worker-20241210184933-192.168.13.129-33159	192.168.13.129:33159	ALIVE	8 (0 Used)	6.7 GiB (0.0 B Used)	
worker-20241210185000-192.168.13.128-40841	192.168.13.128:40841	ALIVE	8 (0 Used)	6.7 GiB (0.0 B Used)	

▾ Running Applications (0)

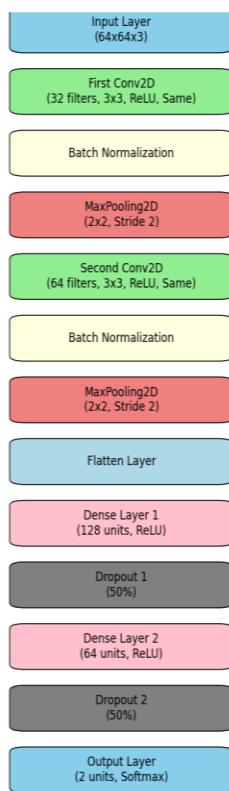
Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
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▾ Completed Applications (1)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
app-20241210185529-0000	Surface Crack Detection	16	4.0 GiB		2024/12/10 18:55:29	root	FINISHED	1.9 min



CNN



Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 64, 64, 3)	0
conv2d (Conv2D)	(None, 64, 64, 32)	896
batch_normalization (BatchNormalization)	(None, 64, 64, 32)	128
max_pooling2d (MaxPooling2D)	(None, 32, 32, 32)	0
conv2d_1 (Conv2D)	(None, 32, 32, 64)	18,496
batch_normalization_1 (BatchNormalization)	(None, 32, 32, 64)	256

conv2d_1 (Conv2D)	(None, 32, 32, 64)	18,496
batch_normalization_1 (BatchNormalization)	(None, 32, 32, 64)	256
max_pooling2d_1 (MaxPooling2D)	(None, 16, 16, 64)	0
Flatten (Flatten)	(None, 16384)	0
dense (Dense)	(None, 128)	2,097,280
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 64)	8,256
dropout_1 (Dropout)	(None, 64)	0
dense_2 (Dense)	(None, 2)	130

Total params: 2,125,442 (8.11 MB)
Trainable params: 2,125,250 (8.11 MB)
Non-trainable params: 192 (768.00 B)

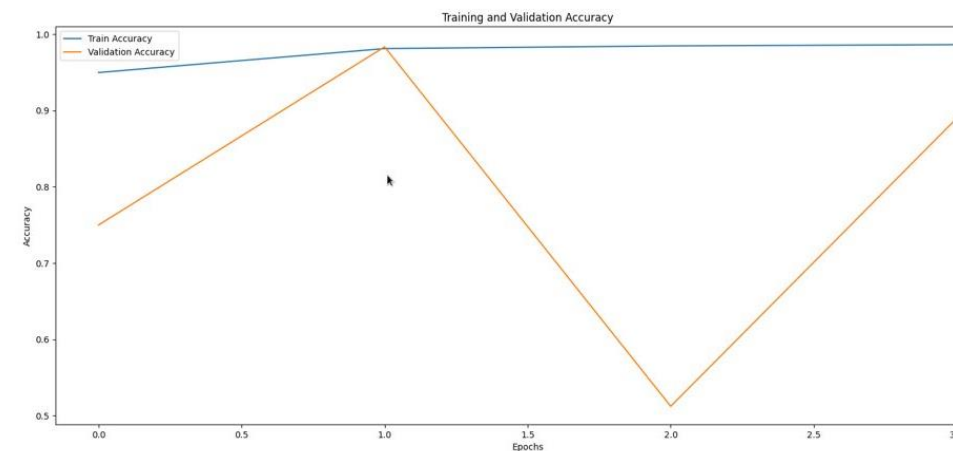
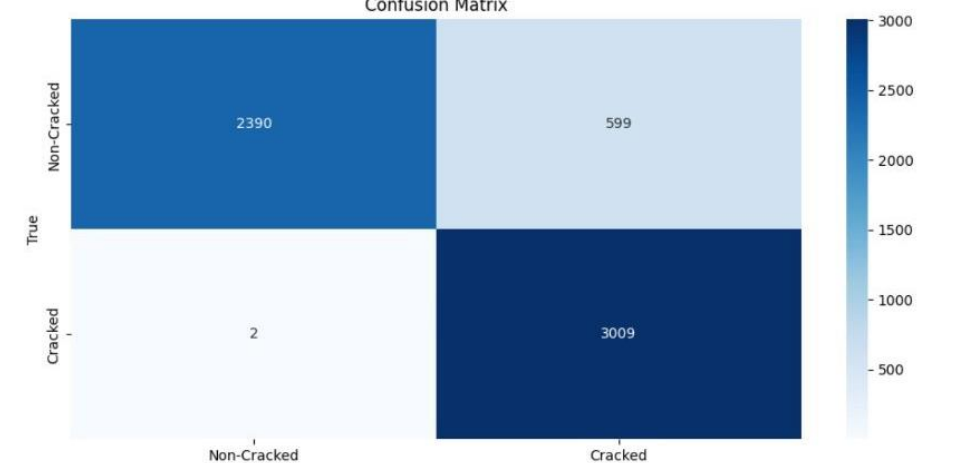
```
usr/local/lib/python3.11/site-packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:121: UserWarning: Your 'PyDataset' class should call 'super().__init__(**kwargs)' in its constructor. '**kwargs' can include 'workers', 'use_multiprocessing', 'max_queue_size'. Do not pass these arguments to 'fit()', as they will be ignored.
  self._warn_if_super_not_called()
```

Hyperparameters

- **Learning Rate:** Default value of the Adam optimizer: **0.001** (not explicitly set in the code).
- **Batch Size:** **16** (used in datagen.flow for training).
- **Epochs:** 4 (number of complete passes through the dataset).
- **Dropout Rate:** **0.5** (used after each dense layer to reduce overfitting).
- This rate specifies the fraction of neurons to drop during training.
- **Optimizer:** **Adam** (used for optimization).

Model Performance:

- The model achieved following results:
- Accuracy: 0.98
- Test accuracy:0.98
- Precision:0.96
- Recall:1.00
- F1 Score:0.98



Classification Report:

	precision	recall	f1-score	sup
Non-Cracked	1.00	0.23	0.38	
Cracked	0.57	1.00	0.72	
accuracy				0.62
macro avg	0.78	0.62	0.55	
weighted avg	0.78	0.62	0.55	

Computational Time

VMRC - VMware Remote Console

Activities Firefox Dec 10 19:20

Spark Master at spark://hadoop1:7077

URL: spark://hadoop1:7077
Alive Workers: 2
Cores in use: 16 Total, 16 Used
Memory in use: 29.2 GiB Total, 8.0 GiB Used
Resources in use:
Applications: 1 Running, 1 Completed
Drivers: 0 Running, 0 Completed
Status: ALIVE

Workers (2)

Worker Id	Address	State	Cores	Memory	Resources
worker-20241210183000-192.168.13.155-46025	192.168.13.155-46025	ALIVE	8 (8 Used)	14.6 GiB (4.0 GiB Used)	
worker-20241210191312-192.168.13.156-35561	192.168.13.156-35561	ALIVE	8 (8 Used)	14.6 GiB (4.0 GiB Used)	

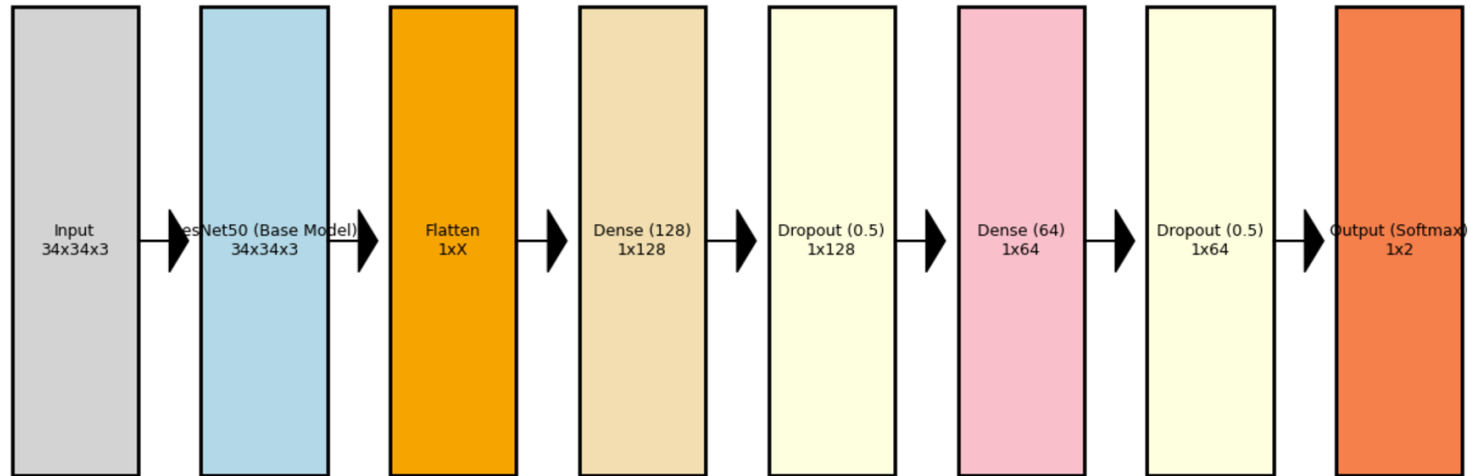
Running Applications (1)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
app-20241210191632-0001	(kill) Surface Crack Detection	16	4.0 GiB		2024/12/10 19:16:32	root	RUNNING	3.2 min

Completed Applications (1)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
app-20241210182736-0000	Surface Crack Detection	0	4.0 GiB		2024/12/10 18:27:36	root	FINISHED	16 s

ResNet50 Architecture



RestNet50

Model: "functional"

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 34, 34, 3)	0	-
conv1_pad (ZeroPadding2D)	(None, 40, 40, 3)	0	input_layer[0][0]
conv1_conv (Conv2D)	(None, 17, 17, 64)	9,472	conv1_pad[0][0]
conv1_bn (BatchNormalization)	(None, 17, 17, 64)	256	conv1_conv[0][0]
conv1_relu (Activation)	(None, 17, 17, 64)	0	conv1_bn[0][0]
pool1_pad (ZeroPadding2D)	(None, 19, 19, 64)	0	conv1_relu[0][0]
pool1_pool (MaxPooling2D)	(None, 9, 9, 64)	0	pool1_pad[0][0]
conv2_block1_1_conv (Conv2D)	(None, 9, 9, 64)	4,160	pool1_pool[0][0]
conv2_block1_1_bn (BatchNormalization)	(None, 9, 9, 64)	256	conv2_block1_1_conv[0...
conv2_block1_1_relu (Activation)	(None, 9, 9, 64)	0	conv2_block1_1_bn[0][...
conv2_block1_2_conv (Conv2D)	(None, 9, 9, 64)	36,928	conv2_block1_1_relu[0...

conv5_block3_2_relu (Activation)	(None, 2, 2, 512)	0	conv5_block3_2_bn[0][...
conv5_block3_3_conv (Conv2D)	(None, 2, 2, 2048)	1,050,624	conv5_block3_2_relu[0...
conv5_block3_3_bn (BatchNormalization)	(None, 2, 2, 2048)	8,192	conv5_block3_3_conv[0...
conv5_block3_add (Add)	(None, 2, 2, 2048)	0	conv5_block2_out[0][0...] conv5_block3_3_bn[0][...
conv5_block3_out (Activation)	(None, 2, 2, 2048)	0	conv5_block3_add[0][0]
flatten (Flatten)	(None, 8192)	0	conv5_block3_out[0][0]
dense (Dense)	(None, 128)	1,048,704	flatten[0][0]
dropout (Dropout)	(None, 128)	0	dense[0][0]
dense_1 (Dense)	(None, 64)	8,256	dropout[0][0]
dropout_1 (Dropout)	(None, 64)	0	dense_1[0][0]
dense_2 (Dense)	(None, 2)	130	dropout_1[0][0]

Total params: 24,644,802 (94.01 MB)
Trainable params: 1,057,090 (4.03 MB)
Non-trainable params: 23,587,712 (89.98 MB)



Michigan Tech

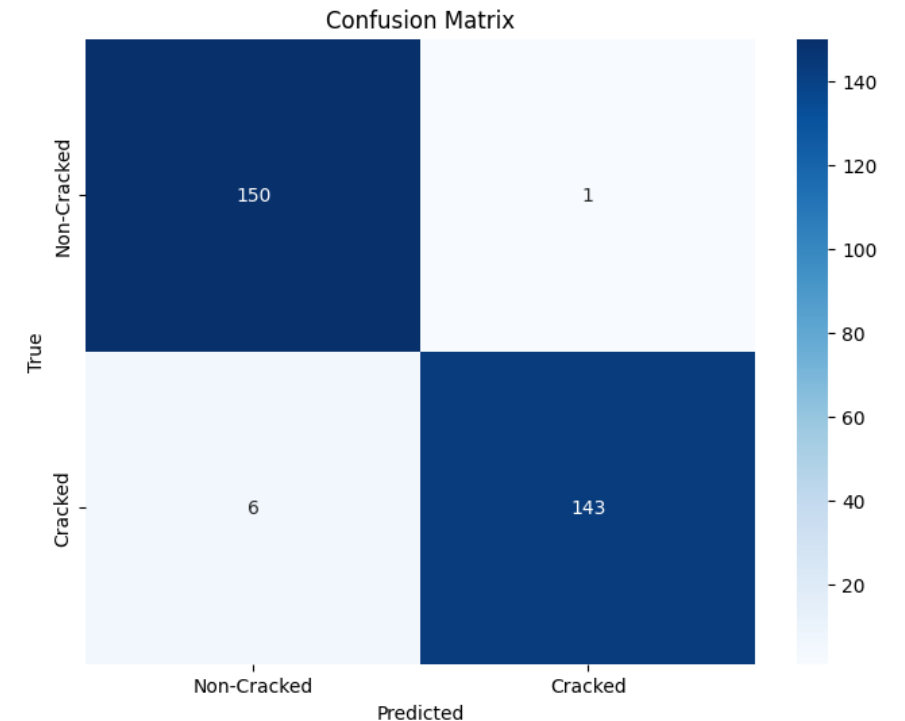
Hyper Parameters

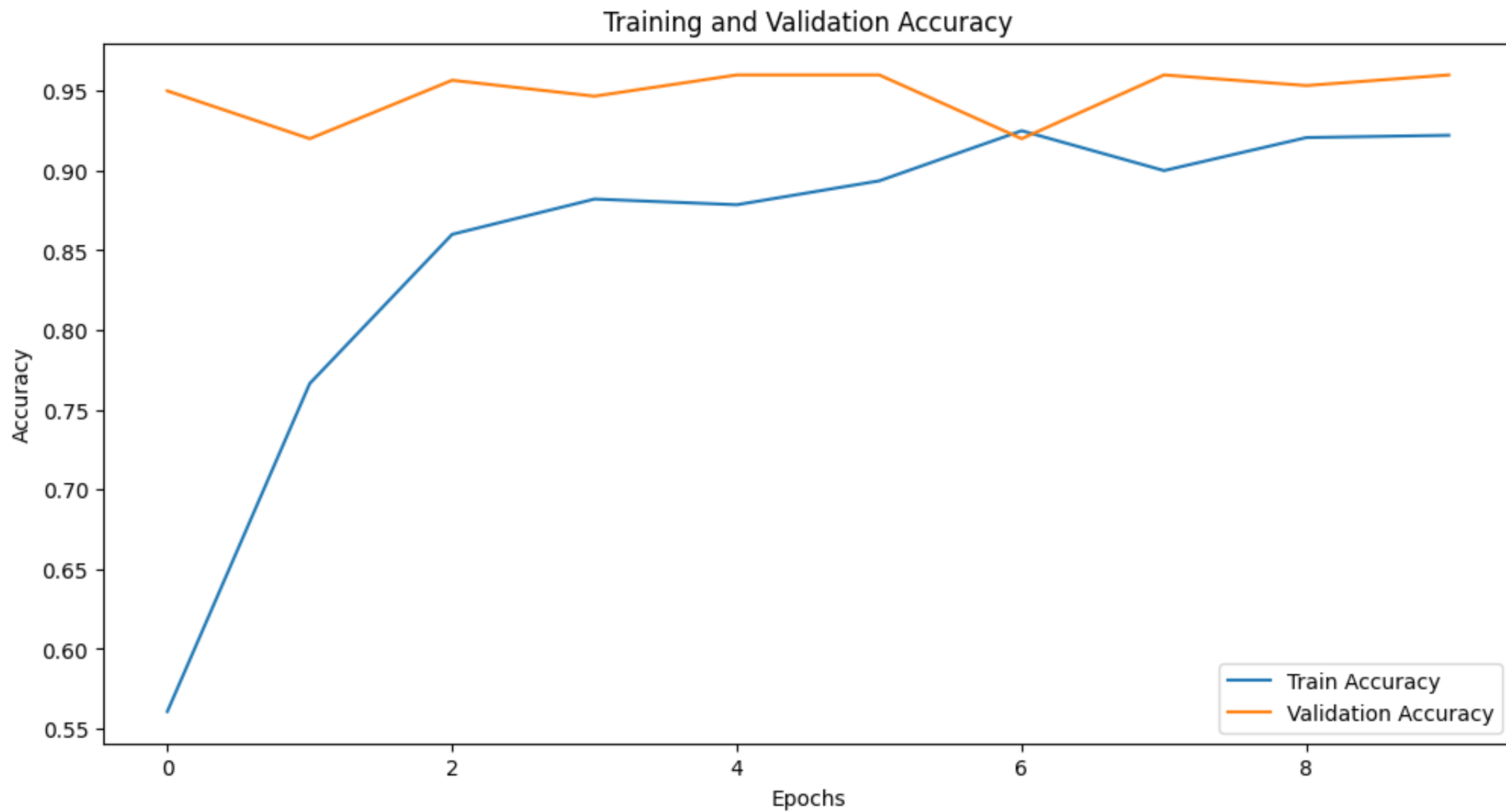
- **Learning Rate:** Default value of the Adam optimizer: **0.001**
- **Batch Size: 16** (a balance between training speed and stability).
- **Epochs: 10** (To balance the computational time).
- **Dropout Rate: 0.5** (used after each dense layer to reduce overfitting).
- **Optimizer: Adam** (used for optimization).



Metrics

Classification Report:				
	precision	recall	f1-score	support
Non-Cracked	0.96	0.99	0.98	151
Cracked	0.99	0.96	0.98	149
accuracy			0.98	300
macro avg	0.98	0.98	0.98	300
weighted avg	0.98	0.98	0.98	300





RestNet50

Computational Time

mmmandyam - hadoop1 - VMware Remote Console

VMRC

Activities

Firefox

Dec 10 19:07

Spark Master at spark://hadoop1:8080

hadoop1:8080

http://hadoop1:9870 http://hadoop1:9864 spark:4040 Spark Master 8080

APACHE SPARK 3.5.0

Spark Master at spark://hadoop1:7077

URL: spark://hadoop1:7077

Alive Workers: 2

Cores in use: 16 Total, 0 Used

Memory in use: 13.5 GiB Total, 0.0 B Used

Resources in use:

Applications: 0 Running, 1 Completed

Drivers: 0 Running, 0 Completed

Status: ALIVE

Workers (2)

Worker Id	Address	State	Cores	Memory	Resources
worker-20241210190133-192.168.13.131-40041	192.168.13.131:40041	ALIVE	8 (0 Used)	6.7 GiB (0.0 B Used)	
worker-20241210190252-192.168.13.132-46503	192.168.13.132:46503	ALIVE	8 (0 Used)	6.7 GiB (0.0 B Used)	

Running Applications (0)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
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Completed Applications (1)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
app-20241210190612-0000	Surface Crack Detection	16	4.0 GiB		2024/12/10 19:06:12	root	FINISHED	23 s



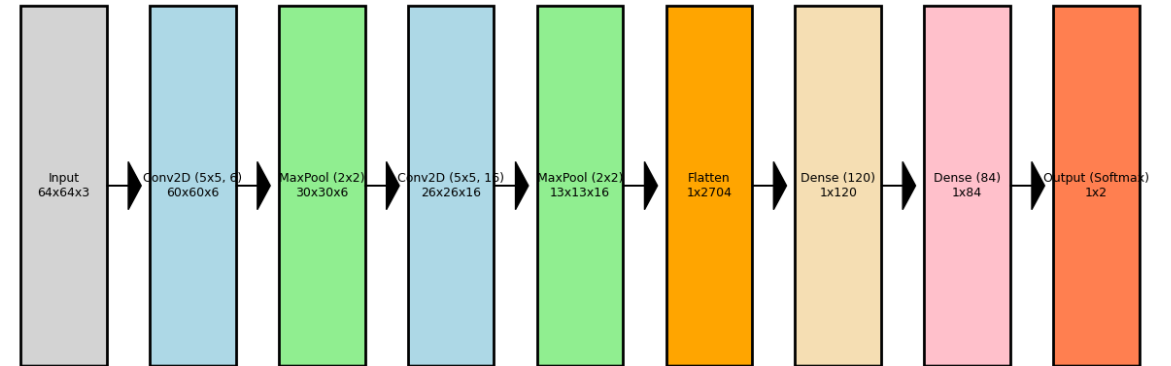
LeNet Model

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 60, 60, 6)	456
max_pooling2d (MaxPooling2D)	(None, 30, 30, 6)	0
conv2d_1 (Conv2D)	(None, 26, 26, 16)	2,416
max_pooling2d_1 (MaxPooling2D)	(None, 13, 13, 16)	0
flatten (Flatten)	(None, 2704)	0
dense (Dense)	(None, 120)	324,600
dense_1 (Dense)	(None, 84)	10,164
dense_2 (Dense)	(None, 2)	170

Total params: 337,806 (1.29 MB)
Trainable params: 337,806 (1.29 MB)
Non-trainable params: 0 (0.00 B)

LeNet Architecture



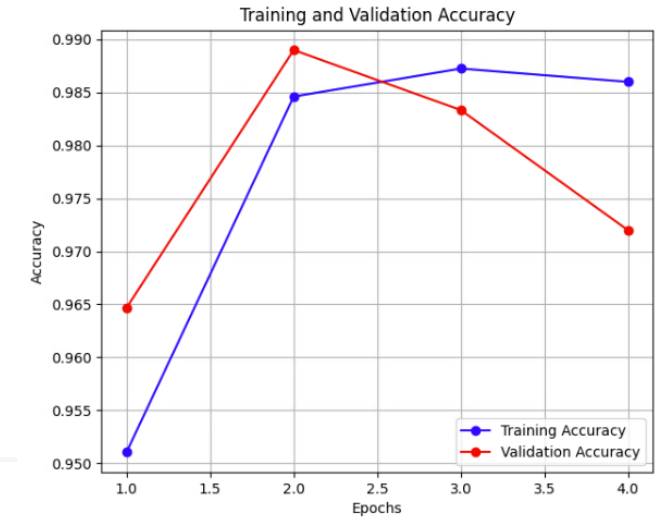
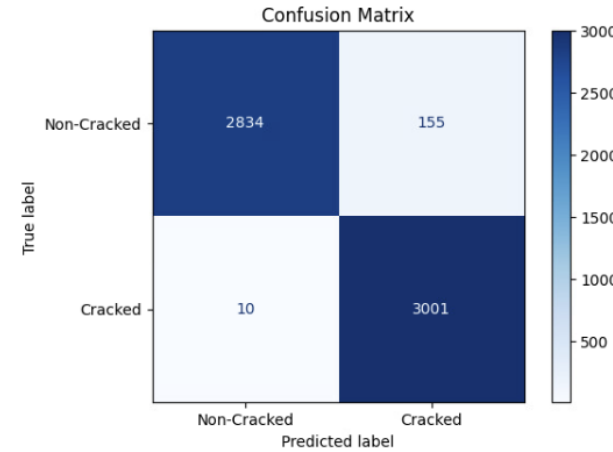
Hyper Parameters

- **Batch Size:** A batch size of 64 is chosen, balancing training time and memory usage for effective learning.
- **Learning Rate:** Default learning rate of the Adam optimizer is used, allowing adaptive learning. Fine-tuning or scheduling could further improve convergence.
- **Epochs:** The model is trained for 4 epochs to balance computational time with performance, though this can be extended for improved results.
- **Optimizer:** The Adam optimizer is employed to leverage adaptive learning rates, ensuring faster and stable convergence during training.

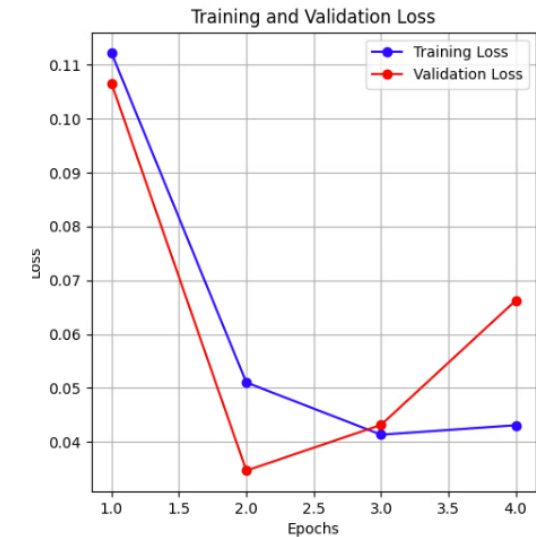


Model Performance:

- The model achieved following results:
- Accuracy: 0.96
- Test accuracy:0.97
- Precision:0.95
- Recall:0.97
- F1 Score:0.97



	precision	recall	f1-score	support
Non-Cracked	1.00	0.96	0.98	2989
Cracked	0.96	1.00	0.98	3011
accuracy			0.98	6000
macro avg	0.98	0.98	0.98	6000
weighted avg	0.98	0.98	0.98	6000



Computational Time

gvannam - hadoop1 - VMware Remote Console

VMRC ▾ || ▾

Dec 9 11:18

Spark Master at spark://h: ×

hadoop1:8080

110% ★

http://hadoop1:9870 http://hadoop1:9864 spark:4040 Spark Master 8080 All Applications

 3.5.0

Spark Master at spark://hadoop1:7077

URL: spark://hadoop1:7077

Alive Workers: 2

Cores in use: 16 Total, 0 Used

Memory in use: 44.9 GiB Total, 0.0 B Used

Resources in use:

Applications: 0 Running, 3 Completed

Drivers: 0 Running, 0 Completed

Status: ALIVE

Workers (2)

Worker Id	Address	State	Cores	Memory	Resources
worker-20241209094246-192.168.13.168-34713	192.168.13.168:34713	ALIVE	8 (0 Used)	14.6 GiB (0.0 B Used)	
worker-20241209094314-192.168.13.167-39557	192.168.13.167:39557	ALIVE	8 (0 Used)	30.3 GiB (0.0 B Used)	

Running Applications (0)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
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Completed Applications (3)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
app-20241209111120-0002	Surface Crack Detection	16	1024.0 MiB		2024/12/09 11:11:20	root	FINISHED	6.5 min
app-20241209105855-0001	Surface Crack Detection	16	1024.0 MiB		2024/12/09 10:58:55	sat3812	FINISHED	13 s
app-20241209095008-0000	Surface Crack Detection	16	1024.0 MiB		2024/12/09 09:50:08	root	FINISHED	10 s

Best Model

- **Best Model: VGG16**
- **Reason:**
 - **Fatal Errors:** Only 1 cracked surface was misclassified as non-cracked.
 - **Metrics:**
 - Recall for cracked surfaces is **0.99**, meaning it identified almost all cracks.
 - Precision is also **0.99**, indicating high reliability.
 - **Confusion Matrix:** The misclassification rate is minimal, with just 1 error for cracked surfaces.
- **Recall for "Cracked":** 0.95 (missed 7 cracked surfaces).
 - **Precision for "Cracked":** 0.95.
 - Overall accuracy: **0.97**.

